Vinay Kumar Chopra

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LeetCode · Codeforces · CodeChef

Education

BITS Pilani **2023 – Present**
Bachelor's in Computer Science *CGPA:* ***8.17*** *India*

Experience

Thriftstone **Jul 2026 – Oct 2026**
AI Engineering Intern *Remote*

- Built a three-tier evaluation framework (extraction, retrieval, agent) for an AI research agent over financial documents, covering **1,925 golden cases across 16 issuers** with zero-tolerance gates for abstention and verification.
- Diagnosed and fixed verifier, ratio-tool, and router defects, including unit-conversion rounding that misgraded exact figures, ratios mishandling missing prior-year balances, and amounts missing units; documented audit findings in evaluation docs.
- Created and extended deterministic agent tools within a **62-tool registry**, including cash-flow ratios and a multi-period cash-flow bridge.
- Collaborated across AI research, data pipelines, retrieval, and deployment, including an AWS ECS feature flag and per-user CloudWatch dashboards.

Open Source Contributions

GRASS GIS (OSGeo) — **4 merged PRs** (Milestones 8.5.0 & 8.6.0)
C, Python, pytest, gprof, GCC/Clang, GitHub

- Parallelized `r.resamp.stats` with OpenMP and per-thread buffers, achieving up to **∼4.7x speedup** with reproducible benchmarks and correctness tests.
- Fixed a maintainer-reported metadata-loss bug in `r.category` that overwrote titles with `rules=`; added pytest regression tests.
- Cleaned up `r.surf.fractal` by replacing a manual initialization loop with `G_zero()`; fixed C11 and security issues, including replacing unsafe `gets()` with `fgets()`.

Projects

SourceMind — Grounded Question Answering over Financial PDFs GitHub
Python, FastAPI, pdfplumber, RapidOCR, sentence-transformers (BGE-small), BM25, Groq Llama 3.3 70B, React

- Built a RAG question-answering system over four HDFC Bank filings that cites the source document and page, and refuses questions unsupported by the filings.
- Implemented PDF ingestion with pdfplumber, RapidOCR fallback for scanned pages, text cleaning, and chunking up to 350 tokens; combined BGE-small dense retrieval with BM25 using Reciprocal Rank Fusion.
- Evaluated 10 answerable, unanswerable, and paraphrased questions: 6 correct answers, 2 correct refusals, 1 wrong answer, and 1 false refusal; traced both failures to terse numeric rows ranking below prose.
- Built a retrieval test harness to check whether the answer-bearing chunk appears in the top 8 and A/B-tested a cross-encoder reranker; recall stayed at 10/17 with or without reranking, and it promoted a wrong figure once, so it remains off by default.

AI Persona Chat — AI Representative GitHub
Next.js, TypeScript, Groq Llama 3.3 70B, Cohere Embeddings and Rerank, Vapi AI, Google Calendar API

- Built a RAG chat and voice assistant over resume and GitHub data using Cohere embeddings and reranking; integrated Google Calendar availability checks and interview-slot booking for Google Meet.
- In manual tests, a relevant chunk appeared in the top 3 reranked results for **9/10** questions; **0/5** out-of-scope questions received invented facts, and first spoken response took about **1.4 seconds** (documented in an evals report).

Aadhaar Seva Radar — UIDAI Data Hackathon GitHub
Python, pandas, NumPy, Matplotlib, Google Colab

- Merged three datasets (**∼4.9M rows**) into a pincode-level table of **31,777 pincodes**; filtered to 3,801 with more than 100 enrolled children aged 5–17 to reduce noise.
- Defined a per-state ratio of biometric updates to enrolments and rule-based zones; flagged Thane as Maharashtra's top red-zone district (12.0x district ratio versus a 20.7x state average) and ranked its pincodes using a weighted priority score.

Technical Skills

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL
AI/NLP & Evaluation: RAG, BGE and Cohere embeddings, BM25, hybrid retrieval, reranking, OCR, retrieval testing, A/B comparison, error analysis
Backend & Web: FastAPI, Spring Boot, REST APIs, Next.js, React, CrewAI, LangGraph
Data, Systems & Tools: Pandas, NumPy, Matplotlib, MySQL, MongoDB, Neo4j, Git, Linux, CMake, Postman, Langfuse, OOP, Networks

Achievements

- **Finalist — Meta PyTorch × OpenEnv Hackathon, Grand Finale** (team of 3): the team built PhonePilot, an RL environment that tests LLM agents for dishonesty (false completion claims, unsupported facts, and sycophancy); I drafted early reward-design ideas adopted by the team.